

	Title,	Year
1	Metal hydrides for solid hydrogen storage: Experimental insights, suitability evaluation, and innovative technical considerations for stationary and mobile applications,	2025
2	Review on thermal design and thermal management for metal hydride reactors: Current status and future development,	2025
3	Use of PCMs in metal hydride reactors for enhancing hydrogen absorption and desorption: A short recent review,	2025
4	Advances in Thermal Management of Integrated Hydrogen Storage Reactors,	2025
5	Enhancing Hydrogen Storage Performance of Metal Hydrides in Ships: A Review,	2025
6	A review on nanofluid, phase change material and machine learning applications for thermal management of hydrogen storage in metal hydrides,	2024

7	A review on research progress in thermal management of metal hydride hydrogen storage reactors; 金属氢化物储氢反应器的热管理研究进展,	2024
8	State of the Art in Development of Heat Exchanger Geometry Optimization and Different Storage Bed Designs of a Metal Hydride Reactor,	2023
9	Thermal Management Techniques in Metal Hydrides for Hydrogen Storage Applications: A Review,	2023
10	Review of thermal management technology for metal hydride reaction beds,	2023
11	Integration of thermal augmentation methods in hydride beds for metal hydride based hydrogen storage systems: Review and recommendation,	2022
12	Review on thermal applications for metal hydrides in fuel cell vehicles: Operation modes, recent developments and crucial design aspects,	2022
13	A design methodology of large-scale metal hydride reactor based on schematization for hydrogen storage,	2022

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Abstract

The application of metal hydrides for hydrogen storage has gained significant attention due to their potential to achieve safe, compact, and efficient storage. This review studies and highlights a need for comprehensive reviews specifically addressing how close certain metal hydrides are to commercialization. It provides a framework for commercial organizations in selecting suitable materials or systems and assist researchers in focusing on unresolved challenges. The review fills that gap by examining the progress and challenges of using metal hydrides as hydrogen storage media, focusing on their suitability for different applications. It categorizes metal hydrides into heavy metal hydrides, such as transition, intermetallic, and high-entropy hydrides, and light metal hydrides. The review evaluates their performance metrics, including gravimetric and volumetric capacities, thermodynamic properties, and kinetics. It compares current metal hydride storage systems with offboard storage, such as refueling stations, and onboard storage for light vehicles. This comprehensively compares current metal hydride storage systems with the remaining gaps to be addressed for commercialization. Advanced thermal management strategies, such as novel reactor designs, are discussed to address the exothermic and endothermic challenges associated with hydrogen absorption and desorption. The review concludes with innovative frameworks for refueling stations and light vehicles. © 2025 Elsevier B.V., All rights reserved.

Metal hydride (MH) hydrogen storage is one of the most important hydrogen storage methods. In MH reactor, the hydrogen storage capacity is directly related to the thermal design of MH reactor is thus particularly important which facilitates rapid heat exchange. Besides, accumulating thermal energy during the hydrogen absorption process, thermal management design is also of great importance on improving the energy efficiency of the hydrogen storage system. In this review, with emphasis on thermal design and thermal management, and the advantages and disadvantages of different heat transfer technologies, the thermal design of MH reactors on the gravimetric and volumetric hydrogen storage density is considered, and the hydrogen storage capacity is analyzed in detail. More importantly, the novel point of this review is the prospect of the future development trend of MH reactor thermal management. The review discusses the reactor and intelligent algorithms and the multiple coupling of MH reactor with other systems such as fuel cells. © 2025 Elsevier B.V., All rights reserved.

With an emphasis on incorporating **phase change materials (PCMs) for better thermal management during hydrogen absorption and release** in the field of hydrogen storage using metal hydride (MH) reactors. Although metal hydrides present potential options for chemical hydrogen storage, their effectiveness is frequently limited by their poor thermal management. By providing controlled heat absorption and release, PCMs in MH reactors offer an efficient thermal management solution for hydrogen storage. The several methods of PCM integration in MH reactors, such as enhanced heat exchangers, fin designs, and material selection, are discussed. Trends in hybridizing PCM-MH systems with other thermal management solutions are also examined, highlighting the potential for improved performance. Research directions are suggested, with an emphasis on improving reactor designs, optimizing PCM properties, and investigating the integration of PCMs in a more efficient and sustainable way. © 2025 Elsevier B.V., All rights reserved.

Metal hydrides, the key to solid-state hydrogen storage, have poor thermal conductivity during hydrogen absorption/release, which limits their effectiveness. Therefore, thermal management technologies are essential to enhance the performance of hydrogen storage reactors. This review discusses the performance of metal hydride hydrogen storage reactors, aiming to provide a theoretical basis for the optimization of thermal management technologies, including fin structures, helically wound tube structures, triply periodic minimal surface (TPMS) structures, and other advanced structures. To enhance the reactor's hydrogen storage and heat transfer efficiency by increasing heat transfer area and optimizing temperature distribution. For example, fin structures have limited heat transfer directions, the production of helically wound tube structures is costly, TPMS structures are complex to manufacture, and fluid viscosity. Comparative analysis clarified the pros and cons of thermal management technologies in heat transfer, hydrogen storage, and reactor design. The proposed future research directions: optimizing fin structures, integrating multiple thermal management technologies, and developing novel TPMS structures. © 2025 Elsevier B.V., All rights reserved.

Some metals and metal alloys can store gaseous hydrogen (GH_2), making the storage of hydrogen in metal hydride (MH) a promising option. This review discusses the marine MHs and technologies for enhancing hydrogen storage performance while identifying future research priorities. First, the review introduces the applications in the maritime field. Subsequently, the technical and economic feasibility of utilizing MHs hydrogen storage is discussed. The review analyzes the scenarios and requirements of ship-based MHs. Furthermore, to meet the target requirements for hydrogen storage properties, the review discusses the hydrogen storage materials, reactor structural parameters, hydrogen storage operating conditions, and thermal management technologies. The hydrogen storage systems and their impacts are summarized. Strategies to enhance the performance of MHs hydrogen storage are proposed. Finally, the review discusses the storage, along with the corresponding thermal management systems, are introduced. Finally, future research directions for ship-based MH hydrogen storage routes for novel MH hydrogen storage materials.

This review is generally about some significant applications related to thermal management of hydrogen storage in metal hydride (MH) reactors. The introductory piece of information about metal hydrides is given and the significance of metal hydride usage in hydrogen storage is discussed. The review discusses the thermal management of hydrogen storage in metal hydrides is given. Later, one of the thermal management technologies, phase change materials (PCMs), is discussed with studies related to the subjects in literature. Finally, different applications of machine learning in thermal management of hydrogen storage have been shown that thermal control for hydrogen storage in metal hydrides may be effectively achieved by employing nanofluids. Generally, Al_2O_3 , CuO , MgO , Fe_2O_3 , GO , GO-SiO_2 , and GO-TiO_2 nanoparticles are used. In metal hydride storage thermal management, the thermal conductivity, heat capacity, and phase change temperature are among the most important parameters that affect the improved performance of the hydrogen storage in metal hydrides. The thermal conductivity, heat capacity, and volume dispersion of PCMs are the fundamental parameters that are investigated in the studies for optimal value. The review discusses the reactor systems, such as supply pressure, inlet velocity, temperature, etc. In order to get a faster response and a more compact reactor system, to a certain degree, the hydrogen storage system benefits from PCM's higher thermal conductivity. The majority of machine learning studies have been used to predict and assess the potential of appropriate materials, metal hydrides, metals, material combinations, etc. for hydrogen storage. The review discusses the machine learning is an effective, realistic means of predicting the parameters influencing hydrogen storage processes without sacrificing accuracy. Future research should take into account more experimental investigations as well as a cost analysis of the

[Objective] Metal hydride hydrogen storage reactors serve as core components in solid-state hydrogen storage techniques. Due to the presence of metal hydrides, significant heat exchange occurs, leading to uneven temperature distribution within the reactors and resulting in inefficient operation of the reactors. Therefore, effective thermal management is recognized as one of the key technologies to improve the efficiency of metal hydride hydrogen storage reactors. This paper aims to review the research progress in the field of thermal management for metal hydride hydrogen storage reactors. A comprehensive literature search was conducted to summarize the principle of hydrogen absorption-desorption reactions in metal hydrides. The heat exchange process was examined, leading to the division of the heat exchange process into three stages: heat transfer in the hydrogen storage material, heat transfer in the metal hydride bed, and heat transfer in the reactor vessel. The efficient operation of a metal hydride reactor depends on the hydrogen sorption and desorption reaction rate. In this regard, the design of metal hydride hydrogen storage systems is crucial. One of the effective solutions for improving the heat and mass transfer efficiency of modern cylindrical-shaped reactors makes it possible to optimize the number of heat exchange elements, design of fins and cooling channels, and metal hydride bed elements. Thus, the development of a metal hydride reactor design with optimal weight and size characteristics, thermal management design, is the relevant task. This paper discusses the influence of different configurations of heat exchangers and metal hydride beds. The advantages and disadvantages of various configurations are considered in terms of heat transfer as well as weight and size characteristics. A comparison of solutions efficiency has been performed, which makes it possible to summarize and facilitate the choice of the reactor configuration.

Metal hydrides are a class of materials that can absorb and release large amounts of hydrogen. They have a wide range of potential applications, such as fuel cells or as a hydrogen release agent for chemical processing. While being a technology that can supersede existing energy storage technologies, there are some challenges that currently hinder their widespread applicability. As the effectiveness of heat transfer across metal hydride beds is a key factor, a description of more efficient heat transfer systems is needed. The literature on the subject has proposed various methods throughout the years, such as optimization of the shape of the reactor vessel, the use of heat exchangers, phase change materials (PCM), and high thermal conductivity additives. This review article provides a comprehensive overview of the latest, state-of-the-art technologies and methodologies and identifies key areas for future researchers to target. A comprehensive analysis of thermal management techniques and various approaches and guidance on selecting appropriate thermal management techniques. For the comparisons, the hydrogen storage capacity and the hydrogen absorbed is studied. This review wishes to examine the various methods that have been used to improve heat transfer efficiency and provide engineers working in the field of hydrogen storage with valuable insights and a roadmap to guide them to further explore the field of metal hydrides. © 2023 Elsevier B.V., All rights reserved.

Metal hydride (MH) hydrogen storage systems have gained interest as an efficient and safe method for hydrogen storage. An understanding of the characteristics of an ideal metal hydride reactor (MHR). However, the thermal conductivity of a metal hydride bed (MHB) is low, which limits its applications and makes it urgent to adopt an efficient thermal management approach. This paper reviews the development of metal hydride hydrogen storage systems and is divided into three parts: (1) from the MHR structure, methods like adding cooling tubes and fins are adopted to enhance the heat transfer efficiency.

This manuscript thoroughly analyzes and comprehensively reviews the reactor configurations and thermal augmentation systems for metal hydride hydrogen storage systems and their issues and challenges in reactor design and its operation. The advantages and issues related to each method are discussed, and the relative system performance on thermal dynamics and hydrogen transfer aspects. The evolution of different types of alloy systems coupled with metal hydride beds are studied carefully. The various conventional and nonconventional heat exchange systems are compared. The heat transfer within and from/to the hydride bed by adding high-performance thermal augmentation systems without a significant weight penalty is an essential factor in reactor design. In conjunction, it is observed that the hydrogen storage technique incorporated should be suitable for various applications. Hence, studies on the gravimetric hydrogen storage capacity of hydrides, reaction kinetics, and heat transfer management are necessary. This article might help analyze the performance characteristics of the hydride reactor and the selection of various thermal augmentation systems for precise applications. © 2022 Elsevier B.V., All rights reserved.

Hydrogen compression for state-of-the-art hydrogen storage systems to pressures of 350 or 700 bar is obviously energy consuming. However, there exists the possibility to transform at least part of this energy into another form of energy leading to an improved overall efficiency. A continuous or discontinuous converter of potential energy into thermal energy. One example is the increase of efficiency for a hydrogen storage system. Hydrogen can contribute to the heating and air conditioning system. With a worldwide steadily growing hydrogen infrastructure, the demand for hydrogen storage systems will increase. Thus, the present manuscript summarizes recent developments in this field and gives detailed recommendations for further research as discontinuous preheating systems or continuous cooling devices. © 2022 Elsevier B.V., All rights reserved.

Hydrogen storage is now the “bottleneck” to realise the application of hydrogen as renewable energy. This is because most metal hydride hydrogen storage reactor (MHR) is the core device used in achieving the desired stable and comprehensive performance of a hydrogen storage system. A methodology and acquired an optimised large-scale MHR (> 50 kg-MHs) for various applications. It mapped and analysed the literature and formed a brief scientometric review from the start. Second, the priority in various applications was concluded in a few keywords. The categories, sketch configurations, and the input/output path of the reaction heat and filling mode, the initial outline of the MHR design, and the outstanding findings. The following optimisation procedure would give designers/investigators a deep understanding of complex thermal management resultant heat and mass transport, pulverisation, self-densification, and stress characteristics were tightly coupled. Subsequently, the system capacity, system gravimetric capacity, system volumetric capacity, and other core dynamic indices. This paper provides a unified design methodology and a roadmap for the design of practical large-scale MHRs and is expected to be satisfied with different application scenarios. © 2022 Elsevier B.V., All rights reserved.

An optimal hydrogen storage reactor should have a higher chemical reaction rate by which the heat can be exchanged as fast as possible. This is a crucial role in improving heat and mass transfer effect in metal hydride beds. Consequently, a variety of different metal hydride bed designs have been proposed and simulated to improve the absorption/desorption rate. In this work, the development of metal hydride bed design is reviewed and optimized to improve the reactor configuration. The summarization and review of metal hydrides design can be broadly classified into five categories: 1) increasing the number of cooling tubes, 2) increasing the length of cooling tubes, 3) increasing and arrangement of cooling tubes, 4) other geometric design, and 5) utilization of phase change material. The effectiveness of various reactor structure configurations is reviewed. The superiority and weakness of different configurations are summarized. A structural parameter analysis for each configuration is also reviewed from the heat and mass transfer aspect. Finally, some reactor designs are proposed. © 2022 Elsevier B.V., All rights reserved.